

BLACK-BOX EXTRACTION · single decode, output only



Document

Black-box LLM

no logits · no gold

Structured output (typed entities + relations)

"entities": [2002 Olympics:MISC, Mediaș:LOC, medal:MISC, China:LOC, IOC:ORG, 1894:TIME, ...]

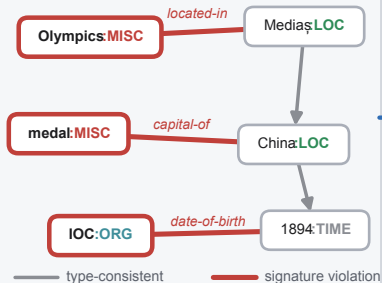
"relations": [located-in, capital-of, date-of-birth, ...]

apply certificate to the output alone

a

Type-signature check

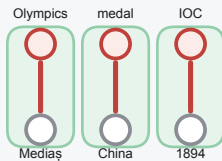
the extractor's own output as a typed graph



b

Conflict graph

each edge: a pair that cannot both be correct



3 vertex-disjoint conflicts

maximum matching = 3

c

Certificate

gold-free · single decode

✓ **≥ 3 items**
provably wrong

sound: every flag is a real error

no gold labels, no training

one deterministic decode

a provable floor, not an estimate